Swiss Institute of Bioinformatics

INTRODUCTION TO SEQUENCING DATA ANALYSIS

Group work

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Adapted from previous year courses

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Projects

Project 1: Variant analysis

Project 2: Long read RNA-seq

Project 3: Short-read RNA-seq

Data analysis steps

Go through all the steps performed in the course:

- Quality control
- Trimming
- Alignment
- Visualization

But also:

- Perform counting for estimating gene expression

Data analysis steps (covered so far)

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Important

Do not <u>only</u> **perform** the calculations, also to **evaluate** the results

Be **reproducible**!!

Bonus exercise

1. Run clumpify.sh after trimming, and the perform alignment: https://www.biostars.org/p/225338/
   - How many reads clumpify.sh removed?
   - Check the alignment rate difference: before and after running clumpify.sh

2. Run Qualimap on aligned files: http://qualimap.conesalab.org/doc_html/analysis.html
   - What extra information do you get?

In the afternoon of day 3, all groups will give a **5 minute presention**

Choose your project

Project 1: Variant analysis

Project 2: Long read RNA-seq

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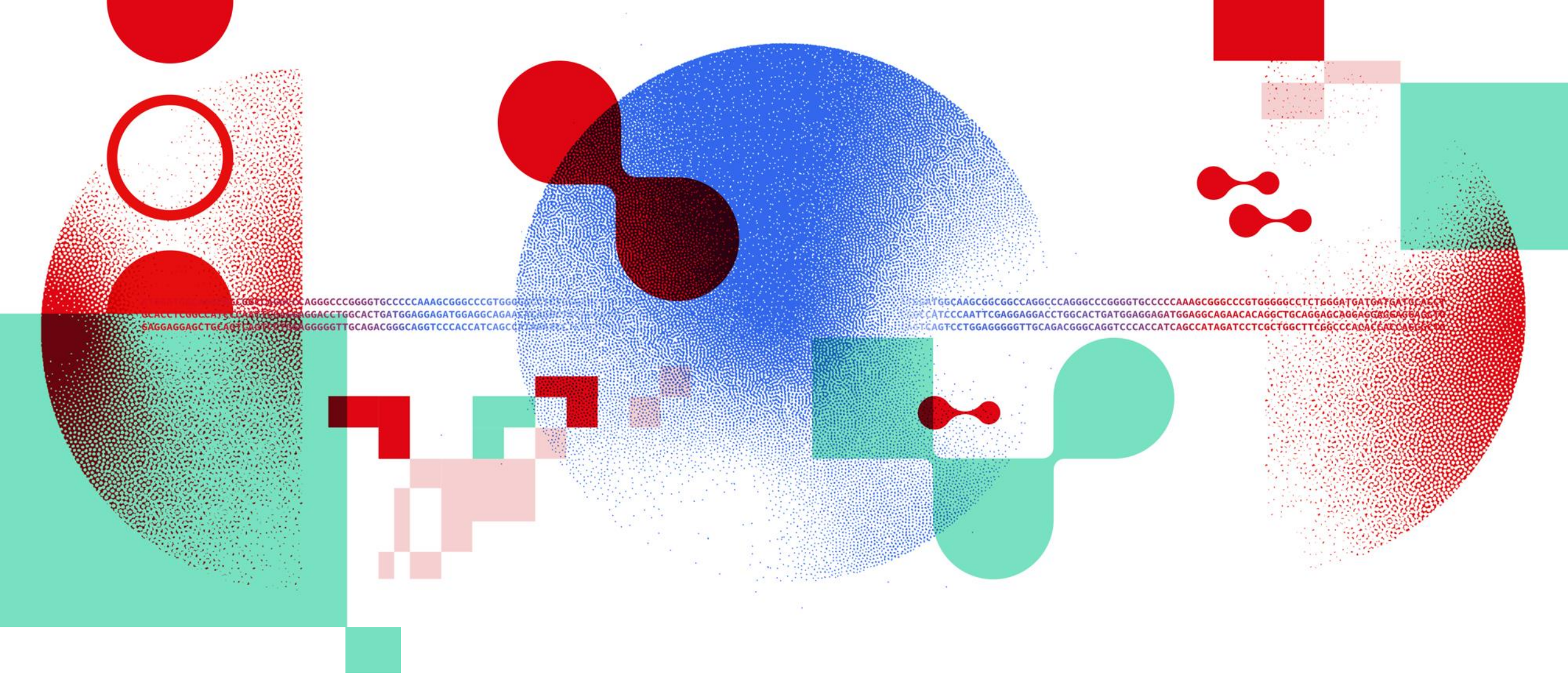
https://forms.office.com/e/tmDa67QVpK



Thank you

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